Improving the Energy Efficiency by Using Quasi-Passive-Dynamics-Based Elastic Actuator

Ruigang Chen, Tongchen Lin, Yizhar Or, and Mingyi Liu

***Abstract*—Electric actuators are commonly used in machines and the optimal energy efficiency can be achieved only at certain speeds. This letter introduces the Quasi-Passive-Dynamics-based Elastic Actuator (QPD-EA), which enhances energy efficiency by leveraging passive dynamics and energy recycling. Instead of directly linking the motor to the robot arm, the motor charges a spring at its optimal energy efficiency speed, and a clutch mechanism controls energy transfer between the spring and the robot arm, enabling efficient movement in any desired direction. The design, modeling, prototype, and experimental validation are described in detail, showing that the QPD-EA consumes 86% less energy than conventional actuators for the same motion, reducing energy use from 154 mJ to 21 mJ. This approach can significantly improve the energy efficiency of actuators in pick-and-place operations.**

***Index Terms*—Actuators, energy efficiency, energy recycling, mechanism design, Quasi-Passive Dynamics.**

I. Introduction

ROBOTS are increasingly replacing humans for dirty, repetitive tasks, with up to 95% of the workload in automobile body shops handled by robots [1]. In the future, robots are expected to take over most physical tasks in factories, buildings, roads, and homes, making them the primary energy consumers. As a result, improving energy efficiency in robots will become crucial. In mobile systems, such as exoskeletons [2], walking robots [3], aerial vehicles [4], and prostheses [5], energy efficiency is a major concern due to the limitations of heavy or bulky energy storage systems like batteries, which restrict operational duration.

In repetitive motion scenarios of actuators, positive work is required to accelerate, while negative work is needed to decelerate, which can contribute significantly to energy consumption. Common methods for deceleration, like mechanical brakes or generate-and-dissipate systems will waste energy—either as heat through braking or as lost energy through resistors. Energy consumption can be greatly reduced by recapturing this negative work, primarily through the following methods.

Recycling negative work with a generator: Negative work can be recycled, where the electric motor acts as a generator, converting negative work into electrical energy for system operations, as seen in electric vehicles [6]. This approach offers controllable energy conversion but is limited by losses in gears [7], [8] electrical-to-mechanical conversion [9], and iron/copper losses [10]. Additionally, batteries cannot absorb sudden energy surges, requiring an ultracapacitor to store and gradually return energy to the battery [11], [12], [13]. Despite these methods, energy losses persist, with a well-designed system achieving a maximum conversion efficiency of 63% [14].

Utilizing potential mechanical energy storage devices: Parallel Elastic Actuators (PEAs) and Series Elastic Actuators (SEAs) demonstrate the potential of using elastic elements for energy recycling (recapturing negative work), which improves robotic efficiency by eliminating iron, copper, and mechanical-to-electrical conversion losses. Springs, with their low hysteresis, serve as effective energy buffers in PEAs. While PEAs are promising for enhancing energy efficiency [15], [16], [17], [18], [19], they cannot independently store or release energy. To address this, Plooij et al. proposed the Bidirectional Clutched Parallel Elastic Actuator (BIC-PEA) [20], but friction and damping in the power transmission limit further efficiency gains. The Switchable Parallel Elastic Actuator (SW-PEA) [21], [22] partially solves this by allowing for energy switching, though the output shaft still cannot operate independently from the motor. Krimsky et al. proposed an elastic energy-recycling actuator [23] based on the PEAs that can partially address this problem. However, the system is not entirely passive hence the energy-efficiency potential is not fully unleashed.

Each motor has its own optimal working speed and most working environments do not match their optimal efficiency point. This situation is inevitable. However, the QPD-EA can solve this problem. The motor can operate at a high-efficiency point at a high speed during the charging process, convert electrical energy into elastic potential energy and store it in the spring. When energy is needed, the clutch mechanism can let the output of the energy in an orderly manner, avoiding the inherent problems of the motor.

In walking robots, there is a branch of research focused on passive dynamics [24], [25], [26], [27]. These robots can start and move steadily by simply being placed on a slope, requiring minimal energy to overcome friction and collision losses, making them highly efficient. Inspired by this, combining actuators

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Ruigang Chen, Tongchen Lin, and Mingyi Liu are with the Department of Mechanical Engineering and Robotics, Guangdong Technion-Israel Institute of Technology, Shantou 515063, China (e-mail: ruigang.chen@gtiit.edu.cn; tongchen.lin@gtiit.edu.cn; mingyi.liu@gtiit.edu.cn).

Yizhar Or is with the Department of Mechanical Engineering, Technion-Israel Institute of Technology, Haifa 3200003, Israel (e-mail: izi@me.technion.ac.il).

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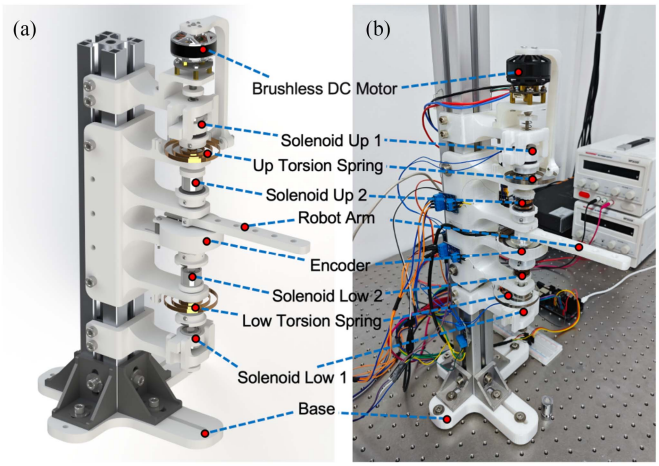


Fig. 1. (a) Design of the QPD-EA. (b) Prototype of the QPD-EA. The QPD-EA mainly consists of two oppositely mounted torsion springs, a robot arm, a BLDC motor, and clutch mechanisms.

TABLE I
COMPARISON OF DIFFERENT ACTUATORS

	BIC-PEA [20]	SW-PEA [21]	QPD-EA
Bidirectional	Yes	Yes	Yes
Lock mechanism	Yes	Yes	Yes
Passive Motion	No	No	Yes
Motor at the optimal velocity	No	No	Yes
Energy saving	53% in realistic task	80% in jumping robots	86% in realistic task

with passive dynamics could further enhance the efficiency. However, current PEAs face limitations, as their motors and springs cannot operate independently, potentially leading to unnecessary energy consumption when attempting to implement passive dynamics.

To address this issue, we propose the Quasi-Passive-Dynamics-based Elastic Actuator (QPD-EA), shown in Fig. 1, which combines passive dynamics and energy recycling to reduce robotic energy consumption. QPD-EA features two reverse-mounted torsion springs above and below the robot arm's output shaft, controlled by a clutch mechanism for efficient energy storage and release. This design allows a brushless DC (BLDC) motor to operate at its optimal speed to achieve the highest efficiency [28], [29], [30] during the charging phase when storing kinetic energy in the springs. QPD-EA can store the kinetic energy of the robot arm in springs at any position, enabling the arm to decelerate to a stop. This stored energy can also be released to accelerate the arm in any direction. During motion, passive dynamics allows the robot arm to swing freely to the target position without motor involvement, reducing energy loss. This approach significantly lowers energy consumption, making it ideal for robots performing rest-to-rest motions.

Table I presents a comparison of actuators similar to QPD-EA, highlighting aspects such as bidirectionality, presence of a locking mechanism, passive motion, motor operation at optimal efficiency, and energy-saving performance.

II. WORKING PRINCIPLE

In this section, the working principle of the QPD-EA will be explained in more detail. First, the components of QPD-EA and how they are connected will be described in detail. Second, the basic principle will be introduced. A horizontal schematic drawing of QPD-EA is shown in Fig. 2, which shows a cart-spring mechanism model for explanation purposes.

A. Clutching Mechanisms

The clutching mechanism is a component that can switch between allowing and preventing relative motion between two components [31]. In Fig. 2, two clutching mechanisms fixed to the ground control the input position x_1 and output position x_2 of the spring at the beginning part. The discrete states of the clutching mechanism are represented by C_1 and C_2. If $C_i = 0$, it indicates that the clutching mechanism locks the spring to the ground, preventing any displacement of this end of the spring. If $C_i = 1$, it means that the spring can move freely. Using this notation, the position of the cart x_c is controllable, and be written in the form

$$x_c = C_2\, x_2 \tag{1}$$

B. Spring

The spring is a low-hysteresis elastic component. In the cart-spring model used to demonstrate the principle, there are two points connected to the input and output of the spring, and the energy stored in the spring is a function of the relative position of these two points. Due to the clutching mechanism introduced in Section II-A, the displacement of the spring is equal to

$$\Delta x = (C_1 \oplus C_2)\,(x_1 - x_2) \tag{2}$$

where the spring is in the equilibrium position when $\Delta x = 0$. Assume that the spring stiffness is large enough, so the displacement caused by compression can be ignored when C_1 and C_2 are equal to 1. The elastic potential energy of the spring is expressed as a function of this relative displacement, as in

$$E_p = f\,(\Delta x) \tag{3}$$

The generalized force $\mathcal{F}$ of the spring at the input and output ends ($i = 1,2$) is defined as

$$\mathcal{F}_i = \frac{\partial E_p}{\partial \Delta x}\,\frac{\partial \Delta x}{\partial x_i} \tag{4}$$

where the direction of the force is defined as the same as the position. In this demonstration example, a compression spring is used. Common springs, such as torsion springs, extension springs, and spiral springs, mostly exhibit a constant positive stiffness characteristic, which means that $\frac{\partial E_p}{\partial \Delta x}$ is a monotonic function of Δx, as in

$$\frac{\partial E_p}{\partial \Delta x} = k\Delta x \tag{5}$$

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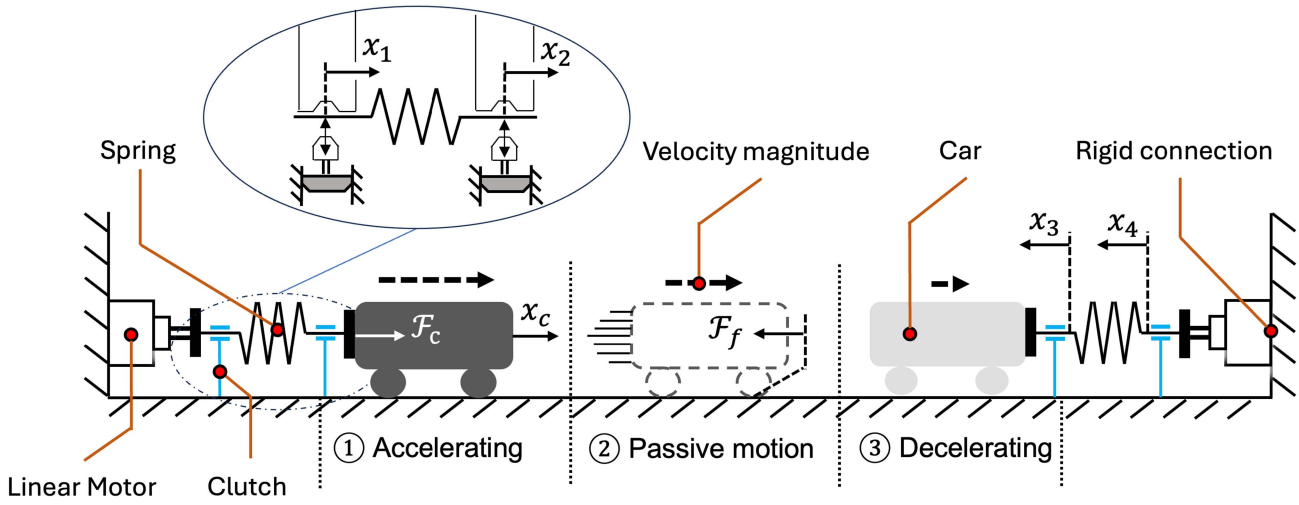


Fig. 2. The QPD-EA concept is visualized as a cart-spring system to explain its basic principles and working mechanisms. The robot arm, represented as the cart, is not always connected to the spring or motor, allowing for passive motion. The magnitude and direction of the elastic potential energy stored in the spring are controlled by the clutching mechanisms, offering flexibility. This visual model is for explanatory purposes and does not represent the final prototype.

where k is the stiffness of the spring that also can be treated as the apparent spring stiffness.

C. Operation Principle

In this demonstration example, there are a total of four clutching mechanisms, with two located at the acceleration terminal and two at the deceleration terminal. During the acceleration phase, the clutching mechanisms at the deceleration terminal do not operate, so there are four states at the acceleration terminal (and vice versa). The focus here is on demonstrating the case where the cart starts from the acceleration terminal and ends at the deceleration terminal.

1) $C_1 = 0$ and $C_2 = 0$: This case allows the moving robot arm to suddenly stop, providing a safety mechanism for future applications. The input and output terminals of the spring are both clutched, so the cart cannot get acceleration from the spring and the motor. If the spring is compressed at this time, it will remain in this compression state.
2) $C_1 = 0$ and $C_2 = 1$: This corresponds to the acceleration and deceleration phase of QPD-EA. The input terminal of the spring is clutched, which means position 1 is locked. Therefore, x_1 now is constant. The compression of the spring depends on the position 2. Combined with (2), the result can be shown in

$$\Delta x = x_{const.1} - x_2 \tag{6}$$

where $x_{const.1}$ is the position of input terminal when x_1 was locked. The generalized force can be shown in

$$\mathcal{F}_2 = \mathcal{F}_c = -\frac{\partial E_p}{\partial \Delta x} \tag{7}$$

where $\mathcal{F}_c$ is the force that the spring applied to the cart. The magnitude and direction of $\mathcal{F}_c$ are the same as $\mathcal{F}_2$.

3) $C_1 = 1$ and $C_2 = 0$: This corresponds to the charging phase of QPD-EA. The output terminal of the spring is clutched, which means position 2 is locked. Therefore x_2 now is constant. The compression of the spring depends on the position 1, as in

$$\Delta x = x_1 - x_{const.2} \tag{8}$$

where $x_{const.2}$ is the position of output terminal when x_2 was locked. The generalized force can be shown in

$$\mathcal{F}_1 = \frac{\partial E_p}{\partial \Delta x} \tag{9}$$

4) $C_1 = 1$ and $C_2 = 1$: This case is similar to SEAs, offering potential for future motor applications. The clutching mechanisms are not functioning, and both the input and output terminals of the spring can move freely. The force from the motor will be transmitted to the cart through the spring.

From (7) and (9), it can be seen that for the same spring, different states of the clutching mechanisms can allow the spring to output force in different directions. Now, let's assume the cart is accelerating. We only need to set the clutching mechanisms at points x_1 and x_2 to $C_1 = 0$ and $C_2 = 1$, respectively, so that the force from the spring can push the cart to accelerate. After a period of passive motion, when entering the deceleration phase, we only need to set the clutching mechanisms at points x_3 and x_4 to $C_3 = 1$ and $C_4 = 0$, respectively, so that the force from the cart can charge the spring, achieving energy recovery.

D. Passive Motion

After the acceleration phase, the cart will be in the passive motion phase. In this phase, the force directly acting on the cart comes from the friction from the ground. In QPD-EA, the moving robot arm will be influenced by the friction due to the contact between itself and the base, as well as the bearings, as in

$$\ddot{x}_c = \frac{\mathcal{F}_f}{m} \tag{10}$$

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where $\mathcal{F}_f$ is the friction from the ground, m is the mass of the cart, $\ddot{x}_c$ is the acceleration of the cart.

Equations (7), (9), and (10) highlight the advantages of QPD-EA, where the torsion spring is not always connected to the motor and robot arm, but instead controlled by a clutch mechanism for engagement and disengagement. This allows QPD-EA to freely control the magnitude and direction of the spring force on the cart. Furthermore, QPD-EA can move passively, influenced only by system friction due to the robot arm is not always connected to the spring and motor. Although friction affects system efficiency, it can be minimized through structural design or external factors like lubricants, thereby improving overall performance.

III. Prototype Design

We built a prototype of QPD-EA, which will be used for energy efficiency comparison tests with conventional high-efficiency motor. In this section, we first discuss the selection of the torsion spring for the prototype. Then, we describe the hardware design of the prototype.

A. Torsion Spring Selection

The spring is a key component of the QPD-EA, as its selection affects both the stored energy and system efficiency. One advantage of the QPD-EA is the ability to control the magnitude and direction of output energy. Here, we define t_a and x_a as the required duration and distance for the task, respectively. We will calculate t_a based on x_a and discuss the selection of spring stiffness accordingly.

For better understanding, the explanation is based on the cart-spring model shown in Fig. 2. The forces acting on the system come from the springs and the friction. The speed of the clutch mechanism is more than an order of magnitude faster compared with the cart. The two springs in the system have already been charged by a motor before the initial phase, with charging distances of x_1 and x_4. The QPD-EA system consists of three stages: ① acceleration, ② passive motion, and ③ deceleration. In both the acceleration and deceleration phases, the initial and final speeds of the springs are zero. The possessed elastic potential energy of the two springs at the beginning of the phase can be expressed by the following equations

$$\begin{cases} E_{p1} = \frac{1}{2}\, k x_1^2 \\ E_{p2} = \frac{1}{2}\, k x_4^2 \end{cases} \tag{11}$$

where E_{p1} and E_{p2} are the initial elastic potential energy of the spring. Note that, when the motor charges the spring, the other end of the spring should be locked by the clutching mechanism.

Considering the potential energy left in the spring is $\frac{1}{2}k(x_1 - x_2)^2$, the terminal speed during the acceleration phase can be expressed by the following equation

$$V_{T1} = \sqrt{\frac{k\left(2x_1x_2 - x_2^2\right) - 2\mathcal{F}_f x_2}{m}} \tag{12}$$

where V_{T1} is the terminal speed of the cart at the end of the acceleration phase.

During the passive motion phase, the system is only subjected to friction. At the end of the passive motion phase, the terminal velocity of the cart is

$$V_{T2} = \sqrt{V_{T1}^2 - \frac{2\mathcal{F}_f x_c}{m}} \tag{13}$$

where V_{T2} is the terminal speed of the cart at the end of the passive motion phase.

QPD-EA experiences the resistance of the spring as well as the friction during the deceleration phase. The energy recycling by the spring can be expressed by the following equation

$$E_r = \frac{1}{2}\, k\left(x_3^2 + 2x_3x_4\right) \tag{14}$$

where E_r is the recycling energy by the spring. The relation between the E_r and V_{T2} can be illustrated by

$$E_r = \frac{1}{2}\, mV_{T2}^2 - \mathcal{F}_f x_3 \tag{15}$$

The distance in the allocated task, x_a, can be represented by the following equation

$$x_a = x_2 + x_c + x_3 \tag{16}$$

Since the duration of the acceleration and deceleration phases is generally very short for most tasks, the average speed can be used to represent the velocity during these phases. The duration in the allocated task, t_a, can be represented by the following equation

$$t_a = \frac{2x_2}{V_{T1}} + \frac{2x_c}{V_{T1} + V_{T2}} + \frac{2x_3}{V_{T2}} \tag{17}$$

x_2 and x_3 are a function of $x_a, t_a, E_{p1}, E_{p2}, m$, and F_f, which are shown as follows

$$[x_2, x_3] = f\left(x_a, t_a, E_{p1}, E_{p2}, m, F_f\right) \tag{18}$$

The x_2 and x_3 can be solved numerically, which controls when is the time the springs engage and disengage with the cart. Those unknowns are solved with Python using scipy.optimize.root function.

Based on the equations presented in this subsection, several variables must be considered when selecting the spring stiffness for QPD-EA. The stiffness not only directly influences the upper limit of the stored elastic potential energy within the system and its overall efficiency but also has a direct impact on the magnitude of V_{T1} when the QPD-EA is under the acceleration phase. This may influence some applications that have specific speed requirements.

Equations from (11) to (17) are based on the cart-spring model, while in QPD-EA, all units should be converted to those used in the rotational system. In addition, m, which represents the mass of the cart, should be expressed as the moment of inertia J. The displacement x needs to be represented as the rotation angle θ.

B. Mechanical Design

In this section, we will introduce the hardware design of QPD-EA (see Fig. 1). Since it is a prototype, primarily designed

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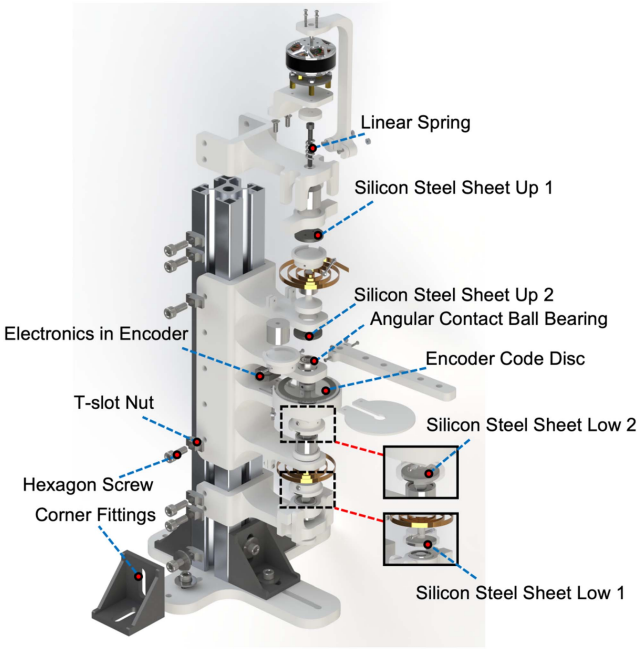


Fig. 3. The exploded view of the QPD-EA. The entire system is fixed to the optical platform using an aluminum profile.

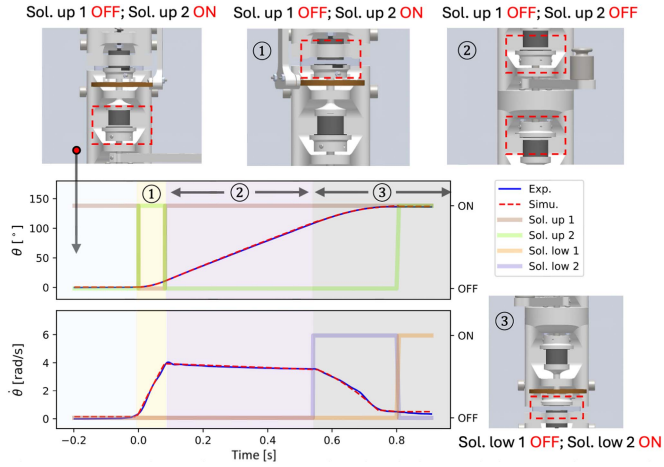


Fig. 4. Comparison chart between the simulation and the experiment. The state of the clutching mechanism is also labeled in the figure. When the solenoid is identified by the red dashed line, it means the solenoid is not functioning at that moment.

to present the concept, the parts of QPD-EA are printed by a 3D printer (Printer model: Hori Z300; Printing material: Polylactic acid) to reduce weight and lower costs. Fig. 3 presents an exploded view of the system. We are going to discuss the clutching mechanism, unique switching mechanism, springs.

The clutching mechanism consists of four solenoids (model: LS-P20/15), each weighing 22.9 g and rated at 1 W, and four circular silicon steel sheets with a diameter of 25 mm and a thickness of 3 mm, containing 3% silicon. The area of the silicon steel sheets is slightly larger than the working surface area of the solenoids to ensure maximum attraction. The solenoids are controlled through the MOSFET circuit by an embedded system via digital signals. The operation of the clutching mechanism has been detailed in Section II-C.

Due to the switching mechanism, the components in QPD-EA are not directly connected to each other but via controlled solenoids. Silicon steel sheets are installed at the endpoints of each component, while solenoids are mounted at the other ends. The switching mechanism utilizes the attraction between the silicon steel sheets and the solenoids when energized to achieve the connection between the components.

For the selection of springs, we use two torsion springs with a stiffness coefficient of 0.01 Nm/rad as the elastic components for energy storage in the system. The outer ends of the torsion springs are fixed to the aluminum profile, while the inner ends pass through a 3D-printed plastic shaft with silicon steel sheets at both ends to facilitate the operation of switching and clutching mechanism.

IV. Simulations

The mass block used in the QPD-EA is a standard cylindrical 50 g weight, fixed at a distance of 108 mm from the center of the rotation axis, simulating a real additional mechanical load for the system. The inertia of the system is $0.00065\ \mathrm{kg}\cdot\mathrm{m}^2$.

Experiments are conducted using the prototype shown in Fig. 1(b). First, the up torsion spring will be deflected 70° by the BLDC to store energy. The up spring will accelerate the robot arm and the mass by a distance of 10° during the acceleration phase. The target angle of the robot arm is set to 135° for this experiment.

The motion state of the proposed system during the acceleration phase can be expressed by the equation

$$\ddot{\theta}(t) = \frac{k\left(\theta(t) + \theta_{start}\right) - F_f}{J} \tag{19}$$

where θ_{start} is the starting angle of the experiment. $\theta(t)$ is the robot arm's displacement versus time. For the passive motion stage, the equation of motion is as follows

$$\ddot{\theta}\ (t) = \frac{F_f}{J} \tag{20}$$

For the deceleration stage, the equation of motion is as follows

$$\ddot{\theta}\ (t) = \frac{-\left(k\theta\,(t) + F_f\right)}{J} \tag{21}$$

The solve_ivp function from the scipy package in Python is used to solve the system of differential equations from the motion equations. The results of solution are shown in Fig. 4, where the displacement θ and velocity $\dot{\theta}$ of the robot arm from the simulation match the experimental result well, indicating that the theoretical basis is correct.

V. Experiment Validation

In this section, an experiment and simulation are designed to compare the energy efficiency of conventional actuators, demonstrating the superior efficiency, and repeatability of QPD-EA.

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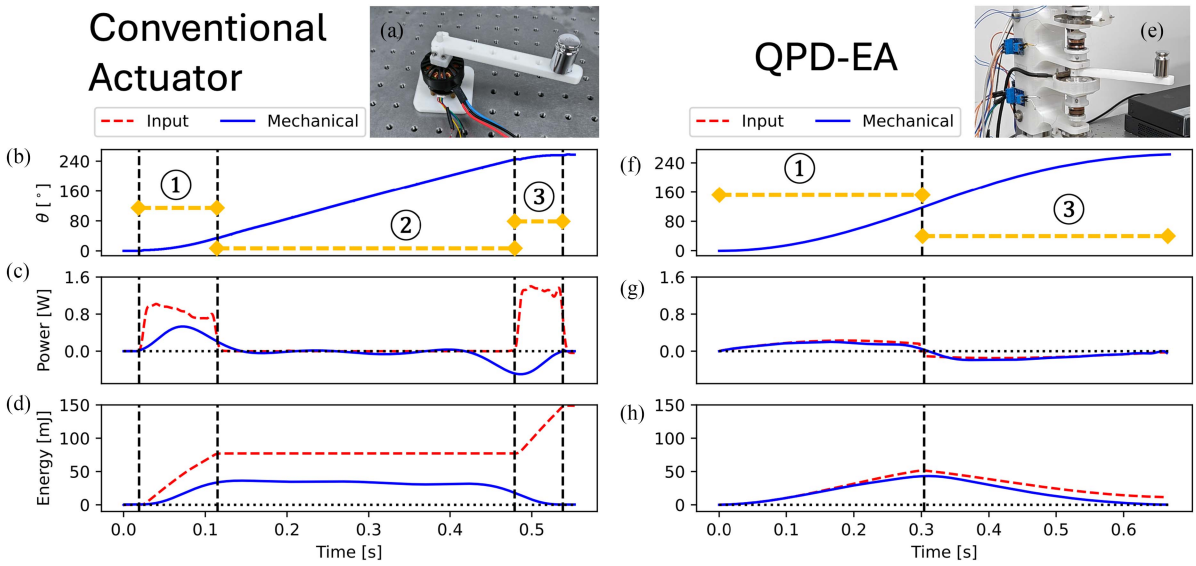


Fig. 5. Displacement, power, and energy analysis: Conventional Actuator vs. QPD-EA. (b)∼(d) represents the displacement, power, and energy curves of the conventional actuator during the motion process. (f)∼(h) represents the displacement, power, and energy curves of QPD-EA during the motion process.

A. Measurement Setup

The experiment on the QPD-EA will be carried out on the prototype shown in Fig. 1(b) The BLDC motor (model: 3508) is mounted above the system and does not directly drive the robot arm. The rotation of the robot arm is measured by a rotary encoder (HKT-56) with a counting rate of 2000 counts per revolution (CPR). To conduct quantitative tests, a 50 g weight is placed on the robot arm to represent rotational inertia. The entire experiment is controlled by an embedded system (OrangePi 5), which also manages the system's state by controlling the MOSFET driving circuits.

The efficiency of the conventional actuator is reflected in the BLDC motor controlled by Field-Oriented-Control (FOC). Field-Oriented Control (FOC) is an advanced method used to control the torque and speed of three-phase electric motors. It provides precise control by aligning the motor's magnetic field with the rotor's field, allowing for efficient operation. We use an open-source FOC controller (Odrive). This controller provides a user interface, and by setting the corresponding parameters, the motor can operate in torque mode, position mode, or speed mode. NI (National Instruments) USB-6003 DAQ is used to measure the voltage and current inputs of the motor, as well as the motor's displacement to calculate the consumed energy and efficiency of the conventional actuator. The three-phase voltage of the motor passes through an RC filter circuit, which is then captured by the USB-6003. The experiment setup is shown in Fig. 6.

B. Comparative Experiments

The comparative experiment highlights the superior energy efficiency of the QPD-EA. The control group uses a conventional actuator (Fig. 5(a)), while the experimental group uses the QPD-EA (Fig. 5(e)), both moving a 50 g weight 260° in 0.6 s. In the experimental group, the spring is deflected 120° by the

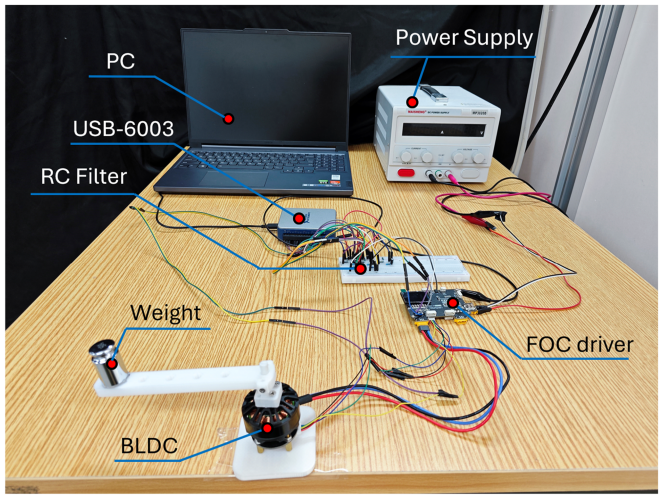


Fig. 6. Energy consumption testing setup for the conventional actuator. The experiment will be conducted on the optical platform to eliminate the effects of gravity.

BLDC motor before the task. During acceleration, the spring aids in moving the weight over 110°. The control group consumes 80 mJ during acceleration and 74 mJ during deceleration, totaling 154 mJ. The experimental group consumes 51 mJ during acceleration and recycles 39 mJ during deceleration, resulting in a net energy consumption of 12 mJ. Considering the efficiency of BLDC as 57% (shown in Fig. 10), 21 mJ of electrical energy is needed to compensate for the 12 mJ of elastic potential energy. QPD-EA will save 86% of energy compared to the conventional actuator.

Both the conventional actuator and QPD-EA can move the weight to the target position. The conventional actuator consumes additional energy during acceleration and deceleration, with a peak power of about 1.2 W. In contrast, the QPD-EA

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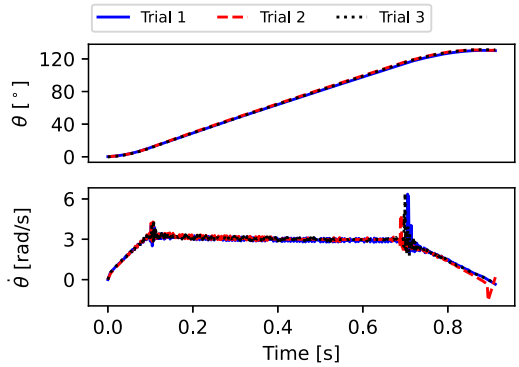


Fig. 7. Repeatability test results. The high overlap rate of the three experimental curves demonstrates the system's repeatability.

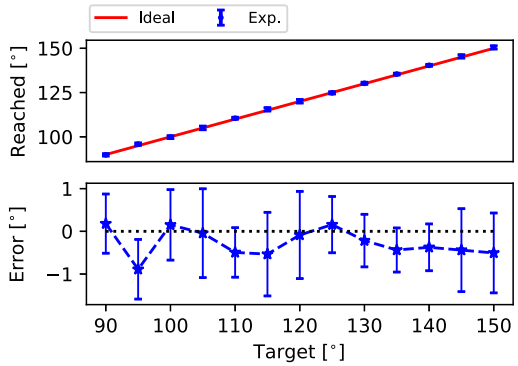


Fig. 8. Accuracy test results. The error between the target angle and the reached angle does not exceed 1°, which demonstrates accuracy.

controls its energy consumption by selectively releasing stored spring energy and recycling the kinetic energy into elastic potential energy during deceleration.

C. Repeatability and Accuracy

The repeatability of the system is one of the important indicators of its application. To test the repeatability of the proposed system, we conducted three experiments on the QPD-EA. The target angle was set to 130°, and the acceleration angle was set to 10°. The results are shown in Fig. 7. It can be observed that the motion trajectories of the three experiments are essentially identical, and the errors in angle and angular velocity are within an acceptable range, demonstrating that the system has repeatability.

To measure the accuracy of the proposed system, experiments were conducted based on the same setup. The target angle of the system started at 90°, with each subsequent set of experiments conducted at 5° intervals, resulting in a total of 13 experimental groups (Angles: 90°, 95°, 100°, …, 145°, 150°).

Each experimental group underwent seven repeated experiments to avoid accidental errors. The experimental results were analyzed using standard deviation. The analysis results are shown in Fig. 8, indicating that the angle of each experimental group deviated from the target angle is no more than 1°, demonstrating that the system possesses accuracy.

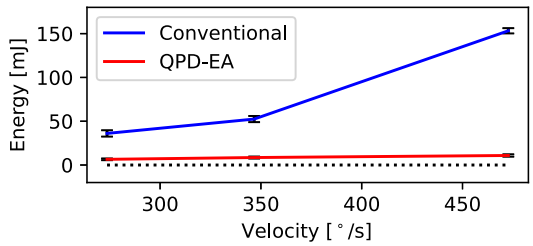


Fig. 9. Energy expenditure test results between the conventional actuator and QPD-EA.

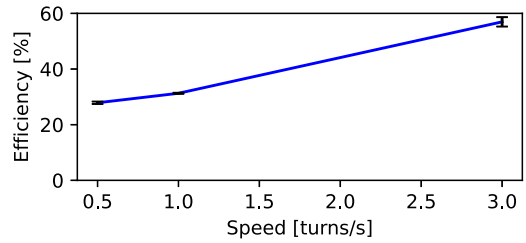


Fig. 10. Results of efficiency of BLDC when charging the torsion spring.

D. Passive Motion and Energy Expenditure

In order to test the efficiency of the system at different speeds, experiments have been carried and the results as shown in Fig. 9. Both the QPD-EA and the conventional actuator are set to reach 260° with three different average speeds. Each set of experiments was performed five times. For the conventional actuator, torque-driven mode is used in the experiment. It is shown that the energy consumption of the conventional actuator increases with the average motion speed. The same motion would require 26∼153 mW of energy, depending on the speed required. However, the energy consumption of the QPD-EA is always around 10 mW despite the motion speed changes. This indicates that unlike the conventional actuator which only have good efficiency at a certain velocity, the QPD-EA can operate efficiently in a broad range of velocities.

The efficiency of charging the spring is a critical factor for power consumption. To measure the energy efficiency of the QPD-EA during the charging phase, a BLDC motor is used to charge the system at a 260°, operating at different speeds. Each experiment is repeated five times to minimize random errors, and results are analyzed with standard deviation, as shown in Fig. 10. It is shown that the efficiency increases with the charging velocity, as expected from the electric motor analysis. The energy efficiency is very low at low motion speed. The QPD-EA's unique switching mechanism enables the decoupling of the motor and the robot arm, which indicates the charging can be done at optimal energy efficiency, despite the robot arm motion requirements. The motor operates at its optimal efficiency during charging, significantly improving overall system efficiency compared to conventional actuators. For example, if the motor runs at 0.5 turns/s, the charging efficiency can only be 28% instead the 57% at 3 turns/s.

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VI. Conclusion

The QPD-EA has the potential to replace many actuators for rest-to-rest motion tasks while reducing energy consumption. The conclusions are summarized as follows:

1) The QPD-EA can reach a predefined arbitrary target angle in desired time with 1° of standard deviation in rest-to-rest motions.
2) The QPD-EA can follow the same motion trajectory with small variations, demonstrating repeatability.
3) After the whole working cycle is completed, compared to the conventional actuator, 86% less energy is consumed by QPD-EA compared to conventional electric actuator while the same motion task is achieved.

However, there is concern that the proposed actuator might be affected by an unexpected obstacle during the free motion phase. This problem can be solved by re-planning the trajectory according to the preset algorithm and its original goal, as presented in the manuscript and the system can get back to its original working cycle again. From the design, the current prototype is assembled by 3D-printed parts, which saves weight but compromises the system's load capacity. It will pose the challenges when dealing with heavy loads. It is feasible to enhance the system's load-bearing capacity by employing a modular, compact design along with robust structural components in the further development. This mechanism can be leveraged in robotics and biological systems to enhance efficiency and enable targets to move further under the same energy consumption.

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